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Proof in School Mathematics: Insights from Psychological Research into Students' Ability for Deductive Reasoning

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There are currently increased efforts to make *proof* central to school mathematics throughout the grades. Yet, realizing this goal is challenging because it requires that students master several abilities. In this article we focus on one such ability, namely, the ability for *deductive reasoning*, and we review psychological research to enhance what is currently known in mathematics education research about this ability in the context of proof and to identify important directions for future research. We first offer a conceptualization of proof, which we use to delineate our focus on deductive reasoning. We then review psychological research on the development of students' ability for deductive reasoning to see what can be said about the ages at which students become able to engage in certain forms of deductive reasoning. Finally, we review two psychological theories of deductive reasoning to offer insights into cognitively guided ways to enhance students' ability for deductive reasoning in the context of proof.

INTRODUCTION

The notion of *proof* has traditionally been associated, almost exclusively, with high school and university mathematics. In recent years, however, many

researchers and curriculum frameworks, especially in the United States, have recommended that proof become central to all students' mathematical experiences throughout the grades (e.g., Ball, Hoyles, Jahnke, & Movshovitz-Hadar, 2002; Hanna, 2000; Hanna & Jahnke, 1996; National Council of Teachers of Mathematics [NCTM], 2000; Schoenfeld, 1994; Yackel & Hanna, 2003). In the research literature, one can find three main reasons for the growing emphasis on proof, even in the early grades. First, proof is fundamental to doing mathematics—it is the basis of mathematical understanding and is essential for developing, establishing, and communicating mathematical knowledge (e.g., Ball & Bass, 2003; Carpenter, Franke, & Levi, 2003; Hanna & Jahnke, 1996). Second, students' proficiency in proof can improve their mathematical proficiency more broadly because proof is “involved in all situations where conclusions are to be reached and decisions to be made” (Fawcett, 1938, p. 120). Third, several researchers (e.g., Ball et al., 2002; Harel & Sowder, 1998; Marrades & Gutiérrez, 2000; Moore, 1994; Usiskin, 1987) have identified students' abrupt introduction to proof in high school as a possible explanation for the many difficulties that secondary school (see, e.g., Chazan, 1993; Coe & Ruthven, 1994; Healy & Hoyles, 2000; Knuth, Choppin, Slaughter, & Sutherland, 2002) and university students (see, e.g., Epp, 1987; Harel & Sowder, 1998; Martin & Harel, 1989; Selden & Selden, 2003; Simon & Blume, 1996; Stylianides, Stylianides, & Philippou, 2004, 2007; Weber, 2001) face with proof, thereby proposing that students engage with proof in a coherent and systematic way throughout their schooling.

Successful engagement with proof requires several abilities by students. One such ability is to recognize that a proof guarantees the truth of a statement for all the elements in the domain covered by the proof (Fischbein, 1982) but does not imply anything about the truth of the statement for elements outside that domain (Stylianides et al., 2007). Another important ability for successful engagement with proof is to recognize the need for a proof. This need may arise from empirical explorations of mathematical phenomena that lead to the formulation of conjectures that are important to a given classroom community (Balacheff, 1990; Ball & Bass, 2003; Boero, Garuti, & Mariotti, 1996; Mason, Burton, & Stacey, 1982). The process of constructing a proof can follow a “zigzag” path between attempts to generate valid arguments and criticisms of these attempts (e.g., through identification of counterexamples; Carpenter et al., 2003; Lakatos, 1976; Lampert, 1990, 1992; Reid, 2002; Schoenfeld, 1983). An important ability required for successful performance in this process is to understand the role of assumptions (e.g., definitions, axioms) that underlie a mathematical argument or a proof (Ball & Bass, 2000; Mariotti & Fischbein, 1997; Stylianides, 2007a; Vinner, 1991; Zaslavsky & Shir, 2005). Finally, one other important ability for successful engagement with proof is to use different forms of reasoning (such as inductive reasoning, reasoning by analogy, and deductive reasoning),

the boundaries among which are often faint.¹ The notion of *deductive reasoning* in particular has been closely related to the notion of proof. From a mathematical standpoint, proof has often been defined in terms of deductive reasoning. For example, Polya (1954), who used the term “demonstrative reasoning” to describe what is now more generally referred to as “deductive reasoning,” noted, “[a] mathematical proof is demonstrative reasoning” (p. v). In addition, from a psychological standpoint, the development of students’ ability for deductive reasoning has been found to go along with the development of their ability for proof (e.g., Foltz, Overton, & Ricco, 1995; Piaget, 1987).

In this article, we focus on one of the abilities for successful engagement with proof that we previously described, namely, the ability for deductive reasoning. Our objective, phrased in general terms, is to review a corpus of psychological research on deductive reasoning to enhance what is currently known in mathematics education research about this ability in the context of proof and to identify important directions for future research.

Available mathematics education research on proof offers: (1) existing evidence that—as early as the elementary grades and in supportive classroom environments—students are able to use deductive reasoning to construct arguments and proofs (e.g., Ball & Bass, 2003; Carpenter et al., 2003; Lampert, 1992; Maher & Martino, 1996; Reid, 2002; Zack, 1997); (2) descriptions of common difficulties that students face in using deductive reasoning in the context of proof (e.g., Chazan, 1993; Galbraith, 1981; Hoyles & Küchemann, 2002; Knuth et al., 2002; Porteous, 1990); and (3) insights into pedagogical factors (including teachers’ instructional practices, knowledge, and beliefs, as well as technological tools and curriculum materials) that play a role in facilitating students’ use of deductive reasoning in the context of proof (e.g., Balacheff, 1991; Ball & Bass, 2003; Boero et al., 1996; Herbst, 2002a; Jones, 2000; Lampert, 1992; Mariotti, Bussi, Boero, Ferri, & Garutti, 1997; Stylianides, 2007b, 2007c; Weber, 2001; Yackel & Cobb, 1996). The findings of this body of mathematics education research, a big part of which is reviewed by Yackel and Hanna (2003) and Harel and Sowder (2007), can be complemented by the findings of psychological research on the development of students’ ability for deductive reasoning. Although this corpus of psychological research makes few connections to the notion of proof, it can offer useful insights into the teaching and learning of proof in school mathematics. Of course, incorporating findings of psychological

¹Each of these forms of reasoning is relevant to most situations of students’ engagement with proof, but there are some situations in which certain forms of reasoning are more relevant than others. For example, inductive reasoning is particularly relevant to generalizing the results of empirical explorations to formulate conjectures, reasoning by analogy is particularly relevant to generating possible ways of constructing a proof based on previous and structurally similar experiences, and deductive reasoning is particularly relevant to constructing proofs or evaluating the validity of given arguments.

studies on deductive reasoning into useful practices for promoting students' ability for proof first requires a considerable amount of research that will aim to build necessary bridges between psychology and mathematics education. This article takes a first step toward addressing this need for research. Specifically, the article aims to address the following questions.

1. Given the findings of psychological research on the development of students' ability for deductive reasoning, what can be said about the ages at which students become able to engage in certain forms of deductive reasoning?
2. Given existing psychological theories of deductive reasoning, what insights can be gained into cognitively guided ways (i.e., ways that consider and are informed by cognitive factors such as working memory capacity) to enhance students' ability for deductive reasoning in the context of proof?

The article is structured into three sections. In the first section, we offer a conceptualization of the meaning of proof in school mathematics and use this conceptualization to delineate our focus on the notion of deductive reasoning. By establishing a conceptual foundation for our review, we are able to synthesize research findings of psychological studies that do not use the same terminology. In the second section, we focus on the first question. In the third section, we focus on the second question and review two psychological theories of deductive reasoning: the *mental models theory* and the *pragmatic reasoning schema theory*. The perspective we took in writing this article was that of two mathematics education researchers who are concerned with issues of proof instruction in school mathematics and who believe that mathematics education research in this domain can be informed by psychological research on deductive reasoning.

THE NOTION OF PROOF AND OUR FOCUS ON DEDUCTIVE REASONING

Despite increased emphasis on the notion of proof in school mathematics, little research has focused on clarifying the meaning of proof in this context, and even less research has attended to this issue in the elementary grades, where proof has traditionally had a limited role (for accounts of the lack of research in this area, see Balacheff, 2002/2004; Reid, 2001, 2005). Stylianides (2007b) proposed a conceptualization for the meaning of proof in school mathematics and used classroom episodes from third grade to elaborate elements of this conceptualization and to illustrate its applicability, even in the elementary grades. He also used this conceptualization to develop a framework for analyzing classroom instruction related to proof to study the role of teachers in cultivating proof among their students. Given our focus in this article on the notion of proof as

early as the elementary grades, this conceptualization of proof is particularly useful. Next, we present the conceptualization and use it to delineate our focus on the notion of deductive reasoning.

The conceptualization of proof is presented in the form of a definition that can be applied in the context of a classroom community at a given time:

Proof is a mathematical argument, a connected sequence of assertions for or against a mathematical claim, with the following characteristics:

1. It uses statements accepted by the classroom community (*set of accepted statements*) that are true and available without further justification;
2. It employs forms of reasoning (*modes of argumentation*) that are valid and known to, or within, the conceptual reach of, the classroom community; and
3. It is communicated with forms of expression (*modes of argument representation*) that are appropriate and known to, or within the conceptual reach of, the classroom community.

(Stylianides, 2007b, p. 291; italics in original)

This conceptualization of proof breaks down each mathematical argument into three major components: the set of accepted statements (e.g., definitions, axioms, theorems), the modes of argumentation (e.g., logical rules of inference such as modus ponens), and the modes of argument representation (e.g., verbal, pictorial, algebraic). Furthermore, the conceptualization seeks to achieve a defensible balance between two (often competing) considerations: mathematics as a discipline and students as mathematical learners. Regarding the consideration of mathematics as a discipline, the conceptualization requires that proofs use true statements, valid modes of argumentation, and appropriate modes of representation. The terms “true,” “valid,” and “appropriate” should be understood in the context of what is typically agreed on in the field of mathematics today. Of course, this is not to say that these terms have universal meaning in the field of mathematics, but it is beyond the scope of this article to elaborate on this issue. Regarding the consideration of students as mathematical learners, the conceptualization requires that proofs depend on what is accepted, known, or conceptually accessible to a classroom community at a given time.

The notion of deductive reasoning corresponds to the component of an argument about modes of argumentation. The conceptualization denotes that the modes of argumentation used in an argument that qualifies as a proof need to be valid and, therefore, need to support logically necessary inferences from a given set of premises. According to commonly accepted notions of deductive reasoning, logically necessary inferences directly implicate the use of deductive reasoning. For example, Kłaczynski and Narasimham (1998) noted that deductive reasoning refers to logically necessary inferences drawn from a “general set of givens or premises” (p. 865). Important to clarify is that we do not associate the

notion of deductive reasoning with particular modes of representation, such as modes that may be characterized as formal versus informal. For example, three classroom communities that use different modes of representation to express their definitions of even and odd numbers—verbal (using everyday language), algebraic, and pictorial—may construct three different proofs for the claim “an odd number plus an odd number equals an even number” as shown in Figure 1. All three proofs are formulated as sequences of assertions that follow logically from the accepted definitions, thereby using deductive reasoning.

In addition to our interest in deductive reasoning as the general form of reasoning associated with logically necessary inferences based on given sets of premises, in this article we pay particular attention to two logically necessary inferences (also known as “logical rules of inference”), which can be considered as two specific forms of deductive reasoning: *modus ponens* (MP) and *modus tollens* (MT). MP states that if the antecedent p of a conditional claim (rule) $p \rightarrow q$ is true, then the consequent q is also true. MP can be summarized as follows: $[(p \rightarrow q) \wedge p] \vdash q$. MT states that if the consequent q of a conditional claim $p \rightarrow q$ is false, then the antecedent p is also false. MT can be summarized as follows: $[(p \rightarrow q) \wedge \sim q] \vdash \sim p$. MP and MT are the mainstay of several proof methods. MP is the foundation of direct proof (illustrated by the three proofs in Figure 1) and the proof method by mathematical induction, whereas MT is the foundation of indirect proof (this includes the proof methods by contradiction and by contraposition).

Given the current emphasis on proof in school mathematics throughout the grades and the connection between deductive reasoning and proof, two important questions arise: Is the ability for deductive reasoning within the conceptual reach of students in school mathematics? And if so, at which ages? Answers to these questions will inform our understanding of the ages at which students can develop aspects of an important ability for proof. To address these questions, we review the findings of psychological research on the development of students’ ability for deductive reasoning.

Proof using everyday language:

Odd numbers are the numbers that if you group them by twos, there’s one left over.

Even numbers are the numbers that if you group them by twos, there’s none left over.

If you add two odd numbers, the two ones that are left over will make another group of two.

The resulting number can be grouped by twos with none left over and, thus, is an even number.

Proof using algebra:

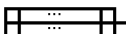
Odd numbers are the numbers of the form $2n + 1$, where n is an integer.

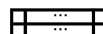
Even numbers are the numbers of the form $2n$, where n is an integer.

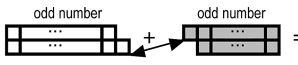
If you add two odd numbers, you get:
 $(2k + 1) + (2m + 1) = (2k + 2m) + (1 + 1)$
 $= 2 \cdot (k + m + 1)$

The resulting number is of the form $2n$ and, thus, is an even number.

Proof using pictures:

Odd numbers are of the form: 

Even numbers are of the form: 

odd number odd number
 =

 even number

FIGURE 1 Three proofs for the claim “odd + odd = even.”

PSYCHOLOGICAL RESEARCH ON THE DEVELOPMENT OF STUDENTS' ABILITY FOR DEDUCTIVE REASONING

Psychological research suggests that students' ability for deductive reasoning develops with age. Specifically, students' ability for deductive reasoning passes through a developmental progression that includes mastery of different forms of deductive reasoning and extends over the entire range of school years (e.g., Byrne & Overton, 1988; Overton, Byrne, & O'Brien, 1985; Ward & Overton, 1990). However, there is no agreement on when mastery of different forms of deductive reasoning is achieved. Some studies suggest that sophisticated mastery of deductive reasoning—including good understanding of the idea of logical necessity—does not emerge before early adolescence (e.g., Markovits, Schleifer, & Fortier, 1989; Overton, Ward, Black, Noveck, & O'Brien, 1987; Piaget, 1928). Some studies suggest that even preschoolers are capable of drawing logically necessary inferences (e.g., Hawkins, Pea, Glick, & Scribner 1984; Richards & Sanderson, 1999). Next we discuss some of these studies.

Piaget's (1928) study on children's mastery of deductive reasoning showed that logically necessary inferences could be observed in children as young as seven or eight years, but the ability for these inferences has certain constraints in these ages. Specifically, it is restricted to empirical reality and deduction is only based on the beliefs that children have adopted themselves. However, from 11–12 years and beyond, a child is capable of what Piaget called *hypothetico-deductive reasoning*; that is, reasoning in which logically necessary inferences are drawn from a set of premises consciously identified as hypothetical. In other words, at the age of 11 or 12, the child becomes, according to Piaget, capable of deductive reasoning based on any assumptions. Markovits et al. (1989) reported findings consistent with those of Piaget. In particular, in a study with 6-, 8-, and 11-year-old children, Markovits et al. showed that even 11-year-olds can find it difficult to distinguish between syllogisms (i.e., arguments from premises to an inference or a conclusion) with premises that are logically connected (e.g., "Every Zoble is yellow. All yellow things have a nose. Do Zoboles have a nose?") and syllogisms with premises that are not logically connected (e.g., "Every Zoble is yellow. All red things have a nose. Do Zoboles have a nose?"). According to Markovits et al., if these children were able to reason deductively with syllogisms, they would be able to recognize situations of undecidability (Piaget, 1987); that is, they would be able to distinguish between premises that are logically connected and those that are not.

Hawkins et al. (1984) gave 4- and 5-year-old-children syllogisms that required logically necessary inferences and were expressed verbally. Many of the children, contrary to Piagetian predictions, could answer a large number of the syllogisms correctly and could justify their responses appropriately. Along similar lines, Richards and Sanderson (1999) suggested that young children possess the

capacity for deductive reasoning with incongruent syllogisms (that is, syllogisms with premises that are in violation of practical knowledge) much earlier than hypothesized by Piaget. An example of such a syllogism is the following: "All fish live in trees. Tot is a fish. Does Tot live in a tree?" In particular, their findings suggested "2-, 3- and 4-year-olds can solve deductive reasoning problems when they are given cues to use their imagination to create an alternative reality where different outcomes are possible" (Richards & Sanderson, 1999, p. B8).

In an effort to address the gaps in the existing research on the development of children's ability for deductive reasoning, Galotti, Komatsu, and Voelz (1997) investigated students' reasoning abilities in kindergarten and grades 2, 4, and 6. They concluded:

[Y]oung children can do more than draw deductive and inductive inferences. Even by second grade, they show the beginnings of implicit recognition that these two types of inferences are different and, as is appropriate, show more consistent answering, and higher confidence, in deductive inferences than in inductive inferences (although confidence in deductive answers is not as high as it should be). (p. 77)

We now turn to findings of psychological studies on the development of students' ability for MP and MT inferences. In a study with early adolescents, middle adolescents, and late adolescents (mean ages around 10, 14, and 17 years, respectively) Klaczynski and Narasimham (1998) concluded, "the rules for drawing MP and MT inferences appear well-developed by early adolescence" (p. 878). This conclusion is consistent with the findings of Osherson's (1974, 1975, 1976) investigations with 10-, 13-, 16-, and 17-year-olds, which showed that the ability for MP is available to 10-year-olds. In fact, Osherson's findings suggested that this ability is available to 10-year-olds, even in the form of multistep deductions. Braine and Rumin's (1983) extensive review of developmental psychology research on students' ability for deductive reasoning showed that the rule for drawing MP inference can be available to even 6-year-olds.

Although performance on MT is usually poorer than performance on MP (Ennis, 1976; O'Brien, 1972, 1975), Falmagne (1980) reported that second-graders can acquire MT, provided that they receive training. Falmagne further reported that training can increase students' capacities to apply MT in different contexts. In addition, as we will see in the next section of the article in which we discuss the mental models theory and the pragmatic reasoning schema theory, it is possible to improve students' performances in different types of deductive reasoning tasks (including tasks that involve MT) by employing several strategies (e.g., by helping students effectively manage their working memory and by recasting formal, abstract problems in everyday, pragmatic contexts).

To conclude, in this section we reviewed psychological research on the development of students' ability for deductive reasoning. This review indicates that

students' ability for deductive reasoning follows a developmental trajectory, wherein some forms of deductive reasoning like MP are mastered earlier than other forms like MT. Even though the findings of existing psychological research do not specify exact ages at which students master different forms of deductive reasoning, all forms of deductive reasoning we reviewed begin to emerge in the early elementary grades. The finding that training can help improve children's ability for MT and increase their capacity to apply MT in different contexts (e.g., Falmagne, 1980) suggests that instruction can help students master even sophisticated (from a cognitive-psychological standpoint) forms of deductive reasoning earlier than the approximate age predicted by the natural developmental trajectory. Indeed, existing evidence from mathematics education research shows that, as early as the elementary grades and in supportive classroom environments, students are able to use deductive reasoning (including forms like MT) to construct arguments and proofs (e.g., Ball & Bass, 2003; Carpenter et al., 2003; Lampert, 1992; Maher & Martino, 1996; Reid, 2002; Stylianides, 2007b; Zack, 1997). Considered together, these psychological and mathematics education findings suggest the important role that mathematics instruction can play in nurturing the development of students' ability for deductive reasoning.

Psychological studies on when students achieve mastery of different forms of deductive reasoning appear to have not paid adequate attention to the role that school mathematics instruction can play in the development of students' ability for deductive reasoning, especially in relation to the teaching of proof. This might explain, at least in part, the lack of convergence in the findings of psychological studies on when students achieve mastery of different forms of deductive reasoning. Future research needs to examine the development of students' ability for deductive reasoning in connection to factors of classroom instruction that can affect this development. In other words, there is a need for an interdisciplinary program of research that will identify the potentially different developmental trajectories of students' ability for deductive reasoning in relation to different instructional conditions. The findings from this research will portray a more exact picture of the ages at which students can engage in certain forms of deductive reasoning under certain instructional conditions, thereby informing also decisions about how to best organize students' engagement with proof.

PSYCHOLOGICAL THEORIES OF DEDUCTIVE REASONING

In this section, we describe and discuss two psychological theories of deductive reasoning: the mental models theory (Johnson-Laird, 1983; Johnson-Laird & Byrne, 1991) and the pragmatic reasoning schema theory (Cheng & Holyoak,

1985; Cheng Holyoak, Nisbett, & Oliver, 1986). These theories provide explanations of people's reasoning processes on certain types of tasks that implicate the use of deductive reasoning, including accounts of the factors that influence people's reasoning on such tasks (e.g., the factors that affect typical errors people make, the factors that support higher performance on certain deductive reasoning tasks than on others). These theories can potentially offer much insight into cognitively guided ways of promoting proof in school mathematics, but their potential has not yet been adequately exploited in the teaching and learning of proof.

Before we describe and discuss the two theories, we explain why these theories are singled out for attention among other rival theories of deductive reasoning, such as the social contract theory (Cosmides, 1989; Cosmides & Tooby, 1989) and the theory of deductive reasoning (Rips, 1994). To explain our choices, we address the more general question: What characteristics should a psychological theory of deductive reasoning have to be a useful theory for mathematics educators to consider? We propose three characteristics that a theory should have.

1. *The theory should have significant empirical support.* There should be strong evidence (from experimental or other kinds of empirical studies) corroborating the tenets of the theory.
2. *The theory should be widely accepted within the field of psychology.* The tenets of the theory and empirical findings that support those tenets should be published in high-quality peer-reviewed psychological research journals, and the theory should be accepted by well-respected researchers other than its founders.
3. *The theory should have been tested with students.* The tenets of the theory should have been tested not only with adults (as it is often the case) but also with students of the level of schooling that is of interest to mathematics educators. For example, for our purposes in this article it is important that a theory has been tested with school students.

Even though it is hard to make these three characteristics operational, we believe that, overall, the two theories we selected for discussion in this article fulfill these three characteristics to a greater extent than other rival theories of deductive reasoning. However, we note that the two theories (like other rival theories) provide an explanation of reasoners' thinking processes on a small range of deductive reasoning tasks, such as syllogistic inference tasks and selection tasks (we say more about these tasks later). These theories have been modified by many researchers to account for other deductive reasoning tasks. For example, English (1998) modified the mental models theory to develop a three-phase theory that provides an explanation of children's deductive reasoning in solving

novel relational problems. In addition, there is still much to be learned about the two theories and how they relate to one another. For example, although there are comparisons of the two theories (see, e.g., Cheng et al., 1986; Moshman, 1998), psychologists have not yet fully analyzed the relationship between these theories.

Finally, we make a comment about the contribution of this section to our goal of taking a step toward building necessary bridges between psychological theories of deductive reasoning and mathematics education research on proof. Ideally, this section would suggest ways to integrate the two theories into a coherent conception of proof instruction in school mathematics. However, the current lack of any linkages between the two theories and mathematics education research on proof has dictated a less ambitious scope for this section. First, we describe each theory and then discuss possible implications of the theory for mathematics education research on proof. We expect that our discussions will lay the ground for the development of more systematic connections between psychological and mathematics education research in the domains of deductive reasoning and proof. The development of these connections will require multidisciplinary research studies in the intersection of psychology and mathematics education.

Mental Models Theory

Description. One of the most influential developments of the past decades in cognitive psychology is Johnson-Laird's (1983; Johnson-Laird & Byrne, 1991) theory of mental models, whereby "mental model" refers to a representation in the mind that has a structure analogous to the structure of the situation it represents. In brief, this theory hypothesizes that the process of deductive thought, as it applies to syllogisms, includes three main stages (Johnson-Laird & Bara, 1984; Johnson-Laird & Byrne, 1991). In the first stage (comprehension stage), the reasoner constructs a mental model of the information presented in the premises of a syllogism. In the second stage (description stage), the reasoner tries to devise a concise description of the model constructed in the first stage that asserts (concludes) something not explicitly stated in the premises. The third stage (validation stage) is where any essential deductive work is carried out. Specifically, in this stage, the reasoner searches for alternative mental models that may lead to refutation of the conclusion drawn in the second stage (counterexamples). If no alternative model is found, then the conclusion is considered valid. If an alternative model is found, however, then the reasoner returns to the description stage to determine whether there is any conclusion that is true in all models constructed up to that point. If such a conclusion is found, then the reasoner searches for alternative models of the premises that may lead to refutation of the conclusion. The search for alternative models continues until all possible models of the premises have been examined. Only a tentative

conclusion can be drawn if there is uncertainty on whether an alternative model of the premises exists. In this theory, developmental changes in the ability for deductive reasoning reflect: (1) improvement of the linguistic competence to comprehend logical terms (e.g., and, or, not, if, none, some, all) in the premises and, accordingly, of the ability to construct appropriate models of those premises; and (2) advancement in the ability to manage these models due to an increase in processing capacity (Johnson-Laird, 1990). Next, we elaborate on important aspects of the theory, and present empirical findings of studies that examined those aspects.

Johnson-Laird and Byrne (1991) argued that “people make deductions by building models and searching for counterexamples” (p. 203).² They considered that the ability for deductive reasoning is equivalent to the “capacity to build models of the world, either directly by perception or indirectly by understanding language, and [the] capacity to search for alternative models” (p. 204). According to the mental models theory, the unfolding of these capabilities occurs under the control of innate constraints.

What develops in childhood is the ability to understand language, the processing capacity of *working memory* (Hitch and Halliday, 1983; Case, 1985), and the meta-ability to reflect on one’s own performance. Seven year-olds cannot cope with syllogisms because they do not understand quantifiers correctly (see Inhelder and Piaget, 1964). Nine year-olds can cope with one-model syllogisms, but not with more than one model (Johnson-Laird, Oakhill, and Bull, 1986; Acredolo and Horobin, 1987). Their *working memory* appears to lack sufficient capacity to retain alternative models of the premises. (Johnson-Laird & Byrne, 1991, p. 204; italics added)

As the quotation suggests, the degree of success in mental model construction and examination depends on a person’s working memory capacity. In other words, the number of models that are constructed and the figural arrangement of terms³ that can be made within the premises, which constitute the two major factors that determine the difficulty of making inferences, seem to be intimately related to working memory capacity.

The effects of both number of models and figure arise from an inevitable bottleneck in the inferential machinery: the processing capacity of working memory, which must hold one representation in a store, while at the same time the relevant information from the current premise is substituted in it. (Johnson-Laird, 1983, p. 115)

²Johnson-Laird and Byrne (1991) used the term *deduction* to refer to *deductive reasoning*. As they noted, in deduction “the goal is to draw a valid consequence from premises” (p. 2).

³For example, syllogisms that involve two premises with three terms (X, Y, and Z) can occur in one of four figures, as shown below.

X - Y	Y - X	X - Y	Y - X
Y - Z	Z - Y	Z - Y	Y - Z

In general, working memory capacity plays a central role in the theory's successful accounting for patterns of performance in deductive reasoning tasks. More specifically, errors occur because limitations in working memory capacity make people fail to consider all possible models of the premises that would provide them with counterexamples to the conclusions they derive from their initial models (Johnson-Laird, 1983; Johnson-Laird & Byrne, 1991). In turn, this limits individuals' ability for validation because it constrains their ability to consider more than one model at a time (Johnson-Laird & Bara, 1984).

Johnson-Laird and colleagues' (1986) experiments with two groups of children (9- to 10-year-olds and 11- to 12-year-olds) supported the claim that the ability to solve syllogistic problems is associated with the number of mental models that have to be constructed for the solution of a given problem. In one experiment in which the two groups of children drew conclusions from 20 pairs of syllogistic premises, no child in either group made a correct response to the three-model problems, whereas all children (the only exception being one 9-year-old) made at least one correct response to a one-model problem. In another experiment in which sixteen 11-year-olds were tested on all 64 possible forms of syllogistic premises,⁴ the children made only 2% correct responses to the three-model problems, as compared with 26% and 63% correct responses to the two- and one-model problems, respectively. Overall, in both experiments, "performance was best with one-model problems and better than chance with two-model problems" (Johnson-Laird et al., 1986, p. 52); correct responses with three-model problems were virtually non-existent.

Investigations with adults revealed a similar pattern of performance; that is, best performance on one-model problems and worst on three-model problems, with the main difference being that the number of problems of each type that the adults solved correctly was typically larger than the corresponding number in children. Interestingly, when adults had only 10 seconds to respond to syllogistic premises, their performance dropped to a level amazingly similar to that of 11-year-olds (Johnson-Laird & Bara, 1984). Anderson, Howe, and Tolmie (1996) made an attempt to explain these results by using Case's (1984) ideas about short-term operating and short-term storage spaces: "while the overall capacity of short-term memory does not increase as a function of development, the effect of practice at tasks results in more efficient use of short-term *operating* space, leaving greater capacity in short-term *storage* space" (Anderson et al., 1996, p. 270; italics in original). Therefore, as learners get older, they become more adept at building, maintaining in memory, and testing a transitory mental model. Increases in the information storage capacity would clearly be beneficial for learners' capacity to achieve these processes.

⁴The 64 possible forms of syllogistic premises are derived as follows: four quantifier combinations for each of two premises that can occur in one of four figures as explained in endnote 3 (i.e., $64 = 4^3$).

Discussion. The mental models theory has not been applied in the teaching and learning of proof, so the examples one can find in the literature illustrating the theory are not focusing on the notion of proof (rather, they are mostly syllogistic tasks placed in non-mathematical contexts). We begin our discussion of the theory with an example of a proving task that we constructed to illustrate possible applications of the theory in the particular domain of proof.

Consider the following two premises, which are basically definitions for multiples of 3 and 6:

An integer is a multiple of 3 if and only if it is three times an integer.

An integer is a multiple of 6 if and only if it is six times an integer.

What can be said (if anything) about any multiple of 6 in relation to a multiple of 3?

Prove your answer.

Using the first premise, the following algebraic expression is constructed for any multiple of 3: $3l$, where l is an integer. Likewise, using the second premise, the following algebraic expression is constructed for any multiple of 6: $6k$, where k is an integer. Using the information in the two premises, the following model is constructed:

Any multiple of 6 is of the form $6k = 3 \cdot (2k) = 3l$, a multiple of 3.

As this model cannot be falsified with the finding of an alternative model, the conclusion is considered valid. That is, we have proved that any multiple of 6 is also a multiple of 3. Successful completion of this proving task depends, given the tenets of the mental models theory, on the linguistic competence of the reasoner to comprehend the logical phrase “if and only if” in the premises and on the reasoner’s processing ability to combine the information from the two premises and search for alternative models.

The proving task we just analyzed was purposefully structured so that there is a clear set of premises from which the reasoner can draw information to construct a mental model. Yet, most of the proving tasks encountered by students and professional mathematicians do not have this clear structure. Typical proving tasks consist only of the question/prompt, and it is the reasoner’s responsibility to select a collection of relevant premises from his or her community’s set of accepted statements (cf. our conceptualization of proof) to construct a proof. In the particular case, the task would likely consist only of the question/prompt: “What can be said (if anything) about any multiple of 6 in relation to a multiple of 3? Prove your answer.” Accordingly, the mental models theory seems to be useful in accounting for the solution of a proving task once a set of premises has been specified (either in the task or by the reasoner). Of course, the reasoner can revise the set of premises by adding or deleting premises to end up with a

sufficient set of premises for the solution of the proving task. Each time a new set of premises is established, the mental models theory can be reapplied.

The mental models theory denotes that limited working memory capacity constrains reasoners' performance in deductive reasoning tasks of which proving tasks are a proper subset. This suggests that mathematics educators can potentially foster the improvement of students' performance in proving tasks in two interrelated ways: (1) by preventing unnecessary usage of students' working memory when they engage with proving tasks, and (2) by helping students develop strategies for effective managing of their working memory capacity.

An example of (1) is for mathematics educators to engage students in "scaffolded" proving tasks (like the one presented earlier) that specify for students a small set of relevant premises for the solution of the task. By excluding irrelevant premises from students' consideration when engaging with a proving task, students are freed from the memory-consuming effort to combine information from more premises and can construct more complicated mental models than they need to. In this way, students are facilitated to focus their attention on the logical structure of the proof and the ideas involved in it. Of course, at some point, educators would like students to become able to identify the relevant premises for the solution of a proving task by themselves. Yet, the types of scaffolded proving tasks described earlier can be useful in the early stages of students' engagement with proof because they can help students develop necessary logical skills that will support less-structured engagement with proof in the future.

An example of (2) is for mathematics educators to help students develop the strategy of "representing information in the premises in equivalent and easier to manage forms" (from a working memory standpoint). Looking back to our analysis of the proving task at the beginning of our discussion of the mental models theory, we reformulated the two premises to algebraic expressions, which, due to their conciseness, reduce the amount of working memory required to process them, thereby facilitating the solution of the task. A related strategy that mathematics educators can help students to develop is "making efficient use of visual records" such as diagrams (see, e.g., Bauer & Johnson-Laird, 1993; English, 1998; English & Halford, 1995; Sweller, van Merriënboer, & Paas, 1998). This strategy can involve not only the creation of a visual record when one is not already available in a proving task but also the use of an existing diagram to practically integrate separate sources of information and to facilitate a solution. It is sometimes the case in proving tasks, especially in geometry, that students are given a diagram (e.g., a geometric figure) and a set of givens, which, although referring to the diagram, are separated from it (Figure 2). To make sense of the two sources of information in Figure 2—the diagram and the givens—young students must mentally integrate them: read the givens, hold them in their working memory, and then search the diagram for the appropriate

Proving task:

A ladybug wants to get from point A to point C following the path in the diagram. How many degrees does the ladybug need to turn to the right at point B? Prove your answer.

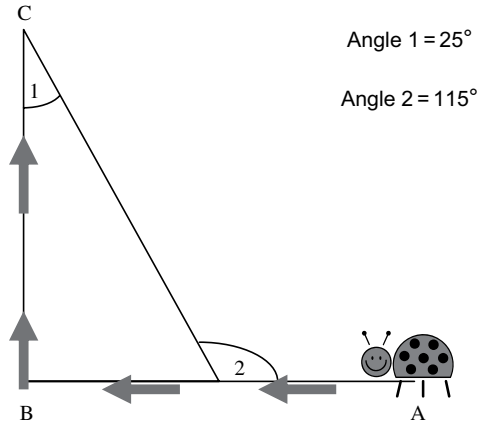


FIGURE 2 A proving task for young students with separate sources of information (the diagram and the givens).

referents. This mental integration process is clearly cognitively demanding and occupies a significant part of students' working memory capacity.

The works of Case (1984) and Anderson et al. (1996) we reviewed earlier suggest that *practice* can help students become more skillful in overcoming the limitations of their working memory capacity, thereby pointing to an educational implication for improving students' ability for deductive reasoning. This implication has first been proposed by Johnson-Laird (1983), based on his observations of "spontaneous improvement in [deductive] reasoning ability simply as a consequence of practice (with no feedback)" (p. 124). Specifically, adults in Johnson-Laird's experiments who have been tested twice within a week showed a 10% improvement in their performance, even without forewarning that they would be retested (Johnson-Laird & Steedman, 1978).

The idea that practice can play an important role in the development of students' ability for deductive reasoning and thus in their ability for proof is not emphasized in the mathematics education literature on proof. Research studies on teaching practices that have successfully promoted students' ability for proof (see, e.g., Ball & Bass, 2003; Carpenter et al., 2003; Lampert, 1990, 1992; Reid, 2002; Stylianides, 2007b; Zack, 1997) do not explicitly identify practice as one of the factors that might have contributed to this success. Part of the reason for which there has been little attention to the potential role of practice in the learning of proof might be that practice has often been associated with secondary aspects

of students' engagement with proof, especially with that of writing proofs in the "two-column form." According to this form, which prevailed in high school geometry courses in the United States for about a century (see, e.g., Herbst, 2002b), "the statements of the proof [are placed] in steps in a column occupying the left half of the page, and . . . the reasons of the statements [are placed] in steps at the right side of the page, with each reason directly opposite its statement" (Shibli, 1932, p. 145). The emphasis on the form made the writing of proofs a ritual procedure that had to be practiced and memorized. As a result, "the substance of proof as a logical and coherent chain of reasoning that guarantees that something must be true became obscured" (Schoenfeld, 1991, p. 325).

Yet, associating practice only with secondary aspects of students' engagement with proof does not do justice to the potential role that practice can play in the learning of proof, as suggested by the psychological research we reviewed in this section. A possible use of practice is to help students develop the strategies for effective management of their working memory capacity (described earlier). In addition, we hypothesize that practice can be used to help students internalize the general logical structure of different proof methods, such as proof by contradiction, thus releasing working memory capacity to be spent in the application of these proof methods. For example, consider the proposition:

There is no smallest positive rational number.

In a proof of this proposition by contradiction, one would start by assuming the opposite of the proposition and would show that it leads to a logical contradiction.

I assume the opposite of the statement I wish to prove: "There is a smallest positive rational number, say y ." Now let $x = y/2$. Then x is a positive rational number that is smaller than y . But this contradicts my initial assumption that y is the *smallest* positive rational number. So I can conclude that the original proposition must be true—"There is no smallest positive rational number."

A student who has not internalized the logical structure of the proof method by contradiction will likely experience increased usage of working memory and, thus, difficulties when trying to apply the method to prove the proposition.

Pragmatic Reasoning Schemas Theory

Description. Cheng and colleagues (Cheng & Holyoak, 1985, 1989; Cheng et al., 1986) were the first to suggest that by evoking certain pragmatic reasoning schemas such as obligation and permission, we can improve university students' performance on certain deductive reasoning tasks generally known as "selection tasks." The pragmatic schemas approach has subsequently been tested with

younger students aged 6 to 15 with equally promising results (e.g., Girotto, Blaye, & Farioli, 1989; Girotto, Gilly, Blaye, & Light, 1989; Girotto, Light, & Colbourn, 1988; Light, Blaye, Gilly, & Girotto, 1989). Before we present the findings of studies in this area, we explain the following terms: pragmatic reasoning schemas (with particular attention to permission and obligation schemas), Wason's selection task, and Reduced Array Selection Task.

Pragmatic reasoning schemas are "clusters of rules that are highly generalized and abstracted but nonetheless defined with respect to classes of goals and types of relationships" (Cheng et al., 1986, p. 294). The set of abstracted rules for situations involving *permission* or *obligation* are two examples of pragmatic reasoning schemas. Specifically, a situation is said to involve permission when "taking a particular action requires satisfaction of a certain precondition" (Cheng & Holyoak, 1985, p. 396). The core of the permission schema can be summarized in the following four production rules.

Rule 1: If action A is to be taken, then precondition B must be satisfied.

Rule 2: If action A is not to be taken, then precondition B need not be satisfied.

Rule 3: If precondition B is satisfied, then action A may be taken.

Rule 4: If precondition B is not satisfied, then action A must not be taken.

According to the pragmatic reasoning schemas theory, if the semantic aspects of a problem suggest to the solvers that they are dealing with a permission situation (typically expressed by the modals "can," "must," "may," and "is required to" and content that is not arbitrary), a permission schema is evoked and the solvers are facilitated to use the entire set of rules that comprise the schema (Cheng & Holyoak, 1985).

An obligation situation is very similar to a permission situation, except from the temporal direction, which is reversed: in a permission situation, execution of an action requires satisfaction of a precondition, whereas in an obligation situation, a certain condition requires performing a subsequent action (Cheng & Holyoak, 1985). Specifically, in obligation situations, "the occurrence of some condition A incurs the necessity of taking some action B" (Cheng et al., 1986, p. 294). The rules about obligations are similar to those about permissions. For example, permission Rule 1 (mentioned earlier) would be phrased as an obligation rule as follows: "If condition A occurs, then obligation B arises." Nevertheless, the inferences one can draw from the two sets of rules are not identical. In particular, an inference from *if A then B* into *A only if B* depends on the reasoning schema to which the *if-then* statement is mapped (Cheng et al., 1986). From an obligation statement of the form, "If condition A occurs, then obligation B arises," one would not want to infer, "Condition A occurs only if obligation B arises." In contrast, from a permission statement of the form, "If action A is to be taken, then precondition B must be satisfied," one can appropriately infer, "Action A is to be taken only if precondition B is satisfied."

The theory of pragmatic reasoning schemas has been tested, almost exclusively, in the context of Wason's (1966) selection task and a simpler version of it—the Reduced Array Selection Task (RAST). Wason's selection task (also referred to as “traditional,” “standard selection task,” or simply “selection task”) is a classic problem in the psychology of deductive reasoning and, as Evans, Newstead, and Byrne (1993) observed, it is “the most intensively researched single problem in the history of the psychology of reasoning” (p. 99).

In the standard selection task, the subjects are shown four cards, each with a letter on one side and an integer on the other side, together with a conditional rule of the form “if p then q .” The rule usually states something like, “If there is a vowel on one side of a card, then there is an even number on the other.” The four cards are laid on the table with each of their upper faces showing one instance of the specific p and q mentioned in the rule; that is, a vowel (p), a consonant ($\sim p$), an even number (q), and an odd number ($\sim q$). The subjects are then asked to indicate all and only those cards that must be turned over in order to decide whether the rule is true or false. Adult subjects usually correctly select the card corresponding to p (equivalent to MP), but, instead of also selecting the logically required $\sim q$ card (equivalent to MT), they usually select the unnecessary q card. The correct selection among adults on this (standard) form of the task is usually about 10% (for a review of related studies, see Evans, 1982).

In the RAST, which was first introduced by Johnson-Laird and Wason (1970), the subjects are also asked to test a rule by the search of potential counterexamples; however, in this task, they are induced to think only about the consequent choices: positive consequent (q) and negative consequent ($\sim q$) of the rule. Specifically, Johnson-Laird and Wason asked adult subjects to inspect shapes of a specified color, one at a time, to verify or falsify given rules with the least number of inspections. One rule they gave to the subjects was, “If they are triangles then they are black,” and, to test it, the subjects could inspect shapes from a box containing black figures and a box containing white figures. Logically, the white box needs to be searched exhaustively and the black box not at all. Instructions to falsify the rule produced better performance (i.e., less black choices) than instructions to verify it. Yet, only just over 50% of the subjects exclusively inspected white shapes when asked to check whether the rule was false (for a review of related studies, see Evans et al., 1993).

According to the pragmatic reasoning schema theory, errors with syllogisms (as defined by the dictates of standard logic) are explained “in terms of the ease to map concrete situations into pragmatic schemas, as well as the degree to which the evoked schemas generate inferences that in fact conform to standard logic” (Cheng et al., 1986, p. 297).⁵ Specifically, the theory denotes that as a

⁵This comes in contrast to the mental models theory, which attributes errors with syllogisms largely to limitations in working memory capacity.

function of their everyday life experiences, people have pragmatic reasoning schemas (such as permission and obligation) that they can use when presented with problems that include semantic cues designed to evoke the schemas. If a schema that produces the same solution to an *if-then* problem (e.g., the standard selection task or the RAST) as does the material conditional is evoked, people are facilitated to solve the problem.

Indeed, in their studies with university students, Cheng and colleagues (Cheng & Holyoak, 1985; Cheng et al., 1986) showed that by evoking permission or obligation schemas the frequency of errors in selection tasks is reduced. Cheng and Holyoak (1985) demonstrated that adults can do significantly better, even in a selection task with an abstract description (i.e., with no reference to any concrete content) of a permission situation than in a syntactically identical version of the standard selection task. Specifically, 61% of the subjects solved the abstract permission problem correctly, whereas only 19% of the subjects solved the standard selection task correctly. Moreover, Cheng et al. (1986) found that offering to subjects a brief training in an obligation schema can improve their performance on selection tasks, even when these tasks do not clearly have the form of obligation rules. In brief, the training included the following:

[The subjects] received a two-page booklet detailing the nature of obligations and the procedures necessary for checking if a violation of the obligation has occurred. An example of an obligation statement presented in the *if-then* conditional form was given. The procedures for assessing obligations were described in terms of four rules, one for each of the four possible situations that might arise (situations that can be mapped onto *p*, *not-p*, *q*, and *not-q*). (p. 310)

Cheng et al. (1986) found that “obligation schema training improved subjects’ performance on problems that were readily interpretable in terms of the appropriate obligation schema” (p. 313). The rule in one such problem was the following: “If a steel support is intended for the roof, then it must be rustproof” (p. 312). They also found that training in the obligation schema improved subjects’ performance on relatively abstract problems that were potentially interpretable in terms of that schema. The rule of one such problem was the following: “If a bolt of cloth has any red threads in it, then it must be stamped with a triangle.”

Studies by Girotto and colleagues (Girotto, Blaye, et al., 1989; Girotto et al., 1988) indicated that evocation of permission and obligation schemas not only facilitates adults’ performance on Wason’s selection task but also facilitates children’s performance on the RAST. Girotto et al. (1988) found that 9- to 10-year-olds were able to solve the RAST, provided that the rule in the test was phrased so that it activated a permission schema. In brief, the children in this study were told a story about bees living in a hive. Then the children were assigned to different conditions. Those in the permission condition were given

the following rule: “All buzzing bees must stay inside in the evenings.” Those in the formal control condition were given the following rule: “All the buzzing bees are inside.” The permission condition elicited 70% globally correct solutions, whereas the corresponding percentage in the control condition was 11%.

In a subsequent study, Girotto, Blaye, and colleagues (1989) obtained similar findings after a computer-based presentation of the same task to subjects that belonged to two different age groups: 9 to 10 years and 14 to 15 years. Specifically, they found that even the youngest children in the sample could solve problems describing unfamiliar permission or obligation rules and that there were no differences in the high rates of correct performance between the two age groups. In addition, their findings suggested, “adolescents and adults experience the same difficulties as young children when the test statements [rules] are not mappable onto a facilitatory [permission or obligation] schema” (Girotto, Blaye, et al., 1989, p. 315). The latter remark is important, for it indicates that the more abstract level of schemas is not a necessary achievement of cognitive development.

The studies we presented thus far indicate that by evoking pragmatic reasoning schemas it is possible to facilitate adults’ performance on a traditional selection task and children’s performance on RAST. Girotto, Gilly, et al. (1989) extended these findings by offering evidence that a context of permission can facilitate the performance of 9- and 10-year-olds, even on the traditional selection task. The following quotation summarizes the findings of various experiments conducted by these researchers with 9- to 10-year-olds and 14- to 15-year-olds to investigate whether young children are able to solve selection tasks that are presented in ways intended to evoke a permission schema.

Experiment 1 demonstrated that both 9–10- and 14–15- year-olds ($n = 160$) could produce a high rate of correct performance in the complete version of the [standard selection] task, both when phrased in terms of a familiar permission and of an explicitly rationalized non-familiar permission. Experiment 2 ($n = 40$) and 3 ($n = 48$) demonstrated that 10-year-olds could solve the task even when the permission rule was not familiar and was plausible only in virtue of the availability of an implicit rationale. (Girotto, Gilly, et al., 1989, p. 79)

One could attribute 9- and 10-year-olds’ high performance on different iterations of the selection task to the fact that children of this age are very close to the stage at which, according to Piaget (1928), ability for deductive reasoning becomes available. To address this issue, Light et al. (1989) conducted a study with 88 six- to eight-year-old children using a form of the RAST. The children were randomly assigned to each of two conditions: the “lorry condition” and the “mushroom condition.” The rule in the lorry condition (“all the lorries must be outside the center”) was contextualized and was intended to represent a plausible and intelligible regulation with which the children did not have

any prior knowledge. The rule in the mushroom condition (“all the mushroom must be outside the center”) was not contextualized and was intended to be an arbitrary rule, albeit one that related to familiar objects and used the exact form of words as the rule in the lorry condition. The children in the lorry condition performed significantly better than the children in the mushroom condition. Performance in the lorry condition reached a nearly 80% success rate among the 7- and 8-year-old children. To conclude, this study demonstrated that even young children, when given a somewhat explicit permission context, can perform highly on the RAST.

Discussion. The findings of studies in the domain of the pragmatic reasoning schema theory suggest that the use of pragmatic reasoning schemas can improve students’ performance on certain deductive reasoning tasks, such as the Wason’s selection task and simpler versions of it (notably, the RAST) in all levels of schooling—from the lower elementary grades to the university level. Specifically, the tenets of the theory, corroborated by strong empirical evidence, denote that evocation of permission or obligation schemas in the context of an *if p then q* task yields significantly fewer errors of all types; that is, errors of avoiding to select $\sim q$ (equivalent to MT), errors of failing to select p (equivalent to MP), and errors of affirming the consequent or denying the antecedent.

The selection task is interesting from a proof standpoint because certain parts of its logical structure are at the core of different proof methods such as proof by contraposition and proof by counterexample. Consider the following two mathematical statements and the corresponding proofs.

Statement 1: If $x^2 \neq y^2$ then $x \neq y$ (where $x, y \in \mathbf{Z}$).

Proof by contraposition: If $x = y$ then $x^2 = y^2$. Hence, the statement is true.

Statement 2: If $x \neq y$ then $x^2 \neq y^2$ (where $x, y \in \mathbf{Z}$).

Proof by counterexample: Consider 1, $-1 \in \mathbf{Z}$. $1 \neq -1$ but $(1)^2 = (-1)^2$. Hence, the statement is false.

If we assume that $x^2 \neq y^2$ maps to p and $x \neq y$ maps to q , then the logical structure of the proof by contraposition for Statement 1 is analogous to selecting $\sim q$ in the selection task. However, an important difference between the reasoning involved in the selection task and the proof by contraposition is that contrary to what happens in the selection task, in proof by contraposition the reasoners do not need to examine what happens in both p and $\sim q$. In other words, once reasoners prove Statement 1 using the proof method by contraposition (beginning with $\sim q$), they do not need to also prove the statement using the direct proof method (beginning with p). To see how Statement 2 corresponds to the selection task, we need to map $x \neq y$ to p and $x^2 \neq y^2$ to q . Then the proof by counterexample

for Statement 2 is analogous to selecting p in the selection task and observing that the rule fails in this case. However, a fundamental difference between the reasoning involved in the selection task and the proof by counterexample is that in the selection task the reasoners are presented with particular cases they can check, whereas in proof by counterexample, the reasoners need to construct the counterexample by themselves. In addition, like in proof by contraposition, in proof by counterexample the reasoners do not need to check what happens in both p and $\sim q$.

The similarities in the logical structure of the selection task and the logical structures of different proof methods, in conjunction with the robust findings of research that selection tasks that evoke a facilitating schema support better performance than selection tasks that do not evoke such a schema, raise the issue of whether it would be fruitful to include selection tasks in the school mathematics curriculum.

On the one hand, mathematics educators can take advantage of the large body of research on how students interact with selection tasks (common difficulties they face, etc.) to organize instructional activities related to deductive reasoning and proof in the context of selection tasks. Instructors can use these activities to help students master logical skills that are useful in different proof methods, notably, proof by contraposition and proof by counterexample. For example, the solution of a selection task that uses the rule *if p then q* can be followed by a discussion of how selecting $\sim q$ relates to MT and also to proof by contraposition. The design and sequencing of the instructional activities on deductive reasoning and proof can draw on the empirically supported tenet of the pragmatic reasoning schema theory that, as a function of their everyday experiences, reasoners have reasoning schemas that can facilitate their performance on selection tasks that include semantic cues designed to evoke the schemas. Thus, depending on the availability or absence of the aforementioned semantic cues in a selection task, mathematics educators can infer the level of difficulty that reasoners are likely to encounter as they try to solve the task. Mathematics educators can also design instructional sequences that begin with selection tasks that provoke students' existing reasoning schemas and continue with selection tasks that do not provoke such schemas and are expected to be more difficult for the students.

On the other hand, mathematics educators need to consider several issues before they can make informed decisions about whether to include selection tasks in the school mathematics curriculum. As we noted earlier, despite the similarities in the logical structure of the selection task and the logical structures of the proof methods of contraposition and counterexample, there are also some important differences between engagement with selection tasks and engagement with the aforementioned proof methods. If we assume that students' engagement with selection tasks is not intended to be a goal of instruction per se but rather a vehicle for enhancing students' ability for deductive reasoning in the context

of proof, then research is needed to understand—from both psychological and teaching standpoints—the relationship between selection tasks and proof methods that involve similar logical structures. For example, how do the cognitive requirements necessary for successful performance on a selection task compare to those necessary for successful performance on a related proving task? And thus, what would it take for instruction to facilitate the transfer of students' ability from solving selection tasks to solving related proving tasks?

The pragmatic reasoning schema theory could have broader implications for instruction if its tenet that schema evocation affects students' performance on selection tasks were shown to apply to other deductive reasoning tasks, including proving tasks. Thus, research is needed to understand the possible effect of evoking pragmatic schemas on students' performance on proving tasks that are placed in pragmatic contexts. Consider, for example, Statement 3 and its alternative formulation, Statement 3', which is embedded in a pragmatic context that aims to provoke an obligation schema.

Original formulation:

Statement 3: If an integer is a multiple of 3, then it is a multiple of 6.

Alternative formulation:

Pragmatic context: A person has marbles each with an integer printed on it. If a marble has a multiple of 6 printed on it, then the person must assign a label "multiple of 6" to the marble.

Statement 3': If a marble has a multiple of 3 printed on it, then the person must assign a label "multiple of 6" to the marble.

We hypothesize that it will be easier for reasoners to prove Statement 3' is false than to prove Statement 3 is false. The rule, "If a marble has a multiple of 6 printed on it, then the person must assign a label 'multiple of 6' to the marble," presumably facilitates the reasoners' use of other rules from the set of rules that comprise the pragmatic schema such as the following: "If a marble does not have a multiple of 6 printed on it, then the person need not assign a label 'multiple of 6' to the marble."

CONCLUSION

In this article, we reviewed a corpus of psychological research on students' ability for deductive reasoning to enhance what is currently known in mathematics education research about this ability in the context of proof. The findings of the psychological research on the development of students' ability for deductive reasoning provide a broad portrait of students' emerging logical skills that can be used to inform educators' understandings of the approximate ages at which

students may be able to master different forms of deductive reasoning that are essential to their engagement with proof. In addition, the two cognitive psychological theories of deductive reasoning offer useful insights into cognitively guided ways to foster students' ability for deductive reasoning in the context of proof. Nevertheless, the translation of research findings of psychological studies into instructional practices that teachers can use to cultivate proof in school mathematics classrooms requires a considerable amount of additional research work.

A major contribution of this article is the identification of several important directions for future research, most of which lie in the intersection of psychology and mathematics education. We will not repeat these directions, rather, we will outline two general themes that cluster/synthesize major directions for future research (which were discussed or alluded to earlier).

1. What is the influence of different instructional conditions on the ages at which different student populations can master certain forms of deductive reasoning (notably, MP and MT)? Corollary questions include:
 - a. What instructional conditions (given specific student populations) can facilitate the mastery of these forms of deductive reasoning?
 - b. How can students' engagement with proof at different levels of schooling be organized based on the forms of deductive reasoning that students can master given favorable instructional conditions?
2. How can the tenets of the mental models and pragmatic reasoning schema theories of deductive reasoning be integrated into a coherent conception of proof instruction in school mathematics? Corollary questions include:
 - a. How do the tenets of the two theories relate to each other, and how do these tenets apply to deductive reasoning tasks (including proving tasks) other than those considered by the two theories?
 - b. How can mathematics instruction address the limitations of students' working memory capacity as they work on proving tasks?
 - c. In what ways can practice support students' learning of proof?
 - d. How do the cognitive requirements necessary for successful performance on syllogistic and selection tasks compare to those necessary for successful performance on related proving tasks? What would it take for instruction to facilitate the transfer of students' ability for solving syllogistic and selection tasks to solving related proving tasks?

An interdisciplinary and collaborative research program that would address the previous questions promises major advancements in relation to not only the practical problem of promoting students' learning of proof but also to the theoretical problem of synthesizing relevant research areas in mathematics education and psychology.

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